EARTH AND PLANETARY SCIENCE LETTERS 3 (1967) 81-88. NORTH-HOLLAND PUBLISHING COMP., AMSTERDAM

THE PALAEOMAGNETIC PROPERTIES OF SOME TERTIARY LAVAS FROM EAST GREENLAND

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Received 21 August 1967

Tertiary lavas from the Scoresby Sund area of East Greenland are found to have moderately stable remanent magnetization which is reversed with respect to the present earth's magnetic field. The mean direction (168,-62) is similar to that of Trap Diabase dykes in S.W. Greenland. The corresponding pole position is slightly different from those of British igneous rocks in the N. Atlantic province but does not suggest continental drifting between Greenland and Europe since the formation of this igneous province. However, further data from Greenland are required to test this dissimilarity before firm conclusions can be drawn.

1. INTRODUCTION

Palaeomagnetic studies in Greenland are particularly interesting as Greenland holds a key position between Europe and North America in any attempted reconstruction of the continents prior to continental drifting. However, few rocks from this area have been studied [1-4] because of the difficulty of access, although some of the earliest palaeomagnetic observations were made on rocks from Greenland [5]. The Tertiary lavas of East Greenland are of particular interest as their probable age, 50-60 m.y., is similar to that of the Tertiary lavas of the other parts of the N. Atlantic province (Faeroes, Scotland, Ireland). Comparison of the directions of remanence between these areas could, therefore, establish if the North Atlantic (Norwegian Sea) had already opened at that time or that the extrusion of the lavas was related to the splitting apart of the earth's crust as continental drift was commencing in this region.

The periodicity of reversals of the earth's magnetic field is also of considerable interest as their zonation is now established for the last 4-5 m.y. from combinations of palaeomagnetic observations and radioactive dates, oceanic cores and sea floor magnetic anomalies [6-8]. However, the establishment of the periodicity of reversals at earlier times is crucial in developing any theory of the origin of the earth's magnetic field or in using palaeomagnetic methods for stratigraphical correlation. Finally, in a succession of stably magnetized lavas, it is possible to determine the magnitude of secular variation [9] as each lava records a spot reading of the direction of the earth's magnetic field and a succession of lavas therefore contains a record of a series of spot readings of the earth's magnetic field during the period of lava eruption. These records yield information on the origin of secular changes and possibly on conductivity variations within the mantle.

In 1965 lavas of Tertiary age were sampled in the Scoresby Sund area by the Oxford University East Greenland Expedition [10] and in 1966 similar lavas were sampled further south by an Oxford University party of the British East Greenland Geological Expedition. These lavas are currently being examined petrologically and petrochemically in the Universities of Oxford and Toronto and only their palaeomagnetic properties are reported here.

2. SAMPLING AND METHODS

Four successions were sampled (fig. 1):

a) Succession 'A' at Pyramiden (70.0°N, 23.1°W) consisting of ten flows dipping 5° to the north.

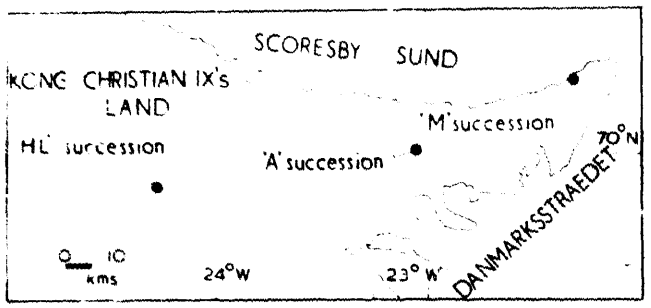


Fig. 1. Locality of sampling sites in the Scoresby Sund area.

b) Succession 'HL' consisting of three slightly weathered lavas from the top of the Kong Christian IX Land basalts at 69.95°N, 28.33°W.

c) Succession 'M' of 24 lavas dipping at 4° to the south at Muslingehjornet (70.1°N, 22.25°W).

d) Succession 'WF' of twelve lavas from the head of Wiedemann's Fjord (68.5°N, 28.33°W, i.e. approximately 300 km S.W. of Scoresby Sund), dipping 5-10° (8° assumed) to the south.

Six samples were collected from each lava (site) in the 'A' and 'WF' successions and up to six samples in the other successions (table 1). Most samples were collected using field drills and orientation was mostly by sun compass. The direction and intensity of remanence of each drill core (sample) was measured with an astatic magnetometer [11, 12] with a reproducibility of better than 1% in intensity and 1° in direction. The susceptibility of each sample was also determined with a susceptibility bridge [13]. The stability of remanence to alternating magnetic field demagnetization was tested in 22 pilot samples (table 2) and the stability index determined [14]. All remaining samples were then treated in 170 peak oersteds in order to reduce the effect of low coercivity components.

3. SUCCESSION 'A'

The initial directions of remanence (table 1) were mostly in a westerly direction and of steep negative (upward) inclination, although the sample directions at two sites (2 and 6) were too scattered for the mean site direction to be significant at a 99% level. Reduction of the low coercivity components improved the scatter of sample directions within most sites, although the mean direction of site 4 was no longer significant. Partial demagnetization of the five pilot samples (table 2) showed metastability in the sample from site 8, and moderate to good stability in the other sites.

All directions of remanence, before and after partial demagnetization as well as the most stable direction isolated in the pilot samples, are all approximately reversed with respect to the present earth's magnetic field direction; there is no evidence of normal polarity. The mean direction after partial demagnetization (table 3) is steeply inclined upwards and just west of south, corresponding to a palaeomagnetic pole position which is significantly displaced from the present south rotational pole.

4. SUCCESSION 'HL'

The initial sample directions at all three sites are widely scattered (table 1) although mostly of southerly declination and upward inclination. Partial demagnetization reduced the scatter of sample directions in each site, although only one site (1, 2, 4) had a significant direction. The single pilot sample from site 7 showed high stability although it came from a site with a large scatter of sample directions.

Despite the large sample direction scatter, the samples appear to be sufficiently stable to indicate that they are all reversely magnetized, although the precise direction of remanence cannot be ascertained in view of the evidence for instability effects in all sites. This instability is probably associated with the slight weathering noted in the samples during collection.

5. SUCCESSION 'M'

The initial remanent directions (table 1) are mostly southerly, but of variable inclination, and five sites do not have significant (99%) mean directions (0, 6, 9, 10, 15). Partial demagnetization reduces the within-site scatter in all sites except one (11) which becomes randomly magnetized but sites 6, 9 and 15 still contain very scattered sample directions.

The pilot samples showed variable behaviour (table 2) with metastability in sites 0, 9 and 11 and moderate stability in the other six samples which were tested. After reduction of the low coercivity compo-

Table 1
Site mean values of remanence.

Site No.	No. samples	Intensity	Susceptibility	Initial			170 (peak) oersteds		
				Decl.	Incl.	c.s.d.°	Decl.	Incl.	c.s.d.°
Succession 'A'									
11	6	60	15	163	−55	13	119	−64	16
10	6	28	19	230	−62	44	310	−77	39
9	6	62	14	222	−80	11	147	−77	18
8	6	11	15	297	−56	25	291	−59	34
7	6	11	73	245	−72	10	210	−66	22
6	6	29	17	251 *	−11 *	68	208	−50	23
5	6	10	17	195	−20	42	210	−58	20
4	6	65	18	238	−59	28	258 *	−64 *	46
3	6	145	22	187	−59	7	189	−61	3
2	6	25	21	220 *	+7 *	79	225	−53	25
Succession 'HL'									
1, 2, 4	3	20	19	141 *	−47 *	81	154	−56	15
5	1	14	16	168 *	−53 *	-	169 *	−60 *	-
6, 7, 8	3	104	18	285 *	−81 *	43	330 *	−82 *	34
Succession 'M'									
15	3	27	17	144 *	−30 *	70	142 *	−42 *	68
13	4	20	17	169	−64	20	200	−80	7
12	5	33	17	197	−69	13	183	−81	6
11	3	18	16	326	+45	12	307 *	−58 *	24
10	6	13	27	204 *	+25 *	77	178	−46	40
9	6	18	27	110 *	−23 *	79	173 *	−71 *	60
8	6	15	13	173	−15	36	175	−27	29
7	5	10	15	192	+14	33	196	−11	31
6	6	7	14	177 *	+72 *	67	120 *	−56 *	49
5	6	43	22	179	−57	38	187	−63	21
4	3	9	33	154	−38	21	157	−43	19
3	6	11	11	138	−29	29	142	−41	13
1	6	29	15	172	−77	44	187	−72	39
0	4	17	28	121 *	+55 *	72	167	+38	25
Succession 'WF'									
L	5	19	7	156	−43	22	141	−46	10
K	6	17	38	157	−76	9	141	−73	12
J	6	12	29	221	0	34	322 *	−73 *	51
I	6	19	28	172 *	−31 *	47	175	−46	26
H	6	27	19	164	−53	4	165	−55	6
G	6	76	39	172	−58	6	173	−60	5
F	6	21	33	192 *	−61 *	64	175	−64	21
E	6	28	22	152	−28	11	148	−29	37
D	6	13	31	185 *	−24 *	56	146	−15	28
C	6	37	44	148 *	+9 *	76	158 *	+18 *	67
B	6	97	30	123 *	−62 *	76	129 *	−62 *	76
A	6	26	26	166	−53	8	161	−55	10

Decl., Incl., are the declination and inclination of the mean remanent vector, before correction for tilt, calculated giving equal weight to each sample vector before and after partial demagnetization in 170 (peak) oersteds. c.s.d.° is the circular standard deviation, containing 63% of the observed directions. The mean intensity and susceptibility are in gauss $\times 10^{-4}$ and gauss oersted$^{-1} \times 10^{-4}$, respectively, and are calculated assuming a log-normal distribution [4].

* Not significant at a 99% level [15, 16]. Each lava within a succession is in stratigraphical order (lowest lava at bottom) but the successions themselves are not necessarily in stratigraphical order.

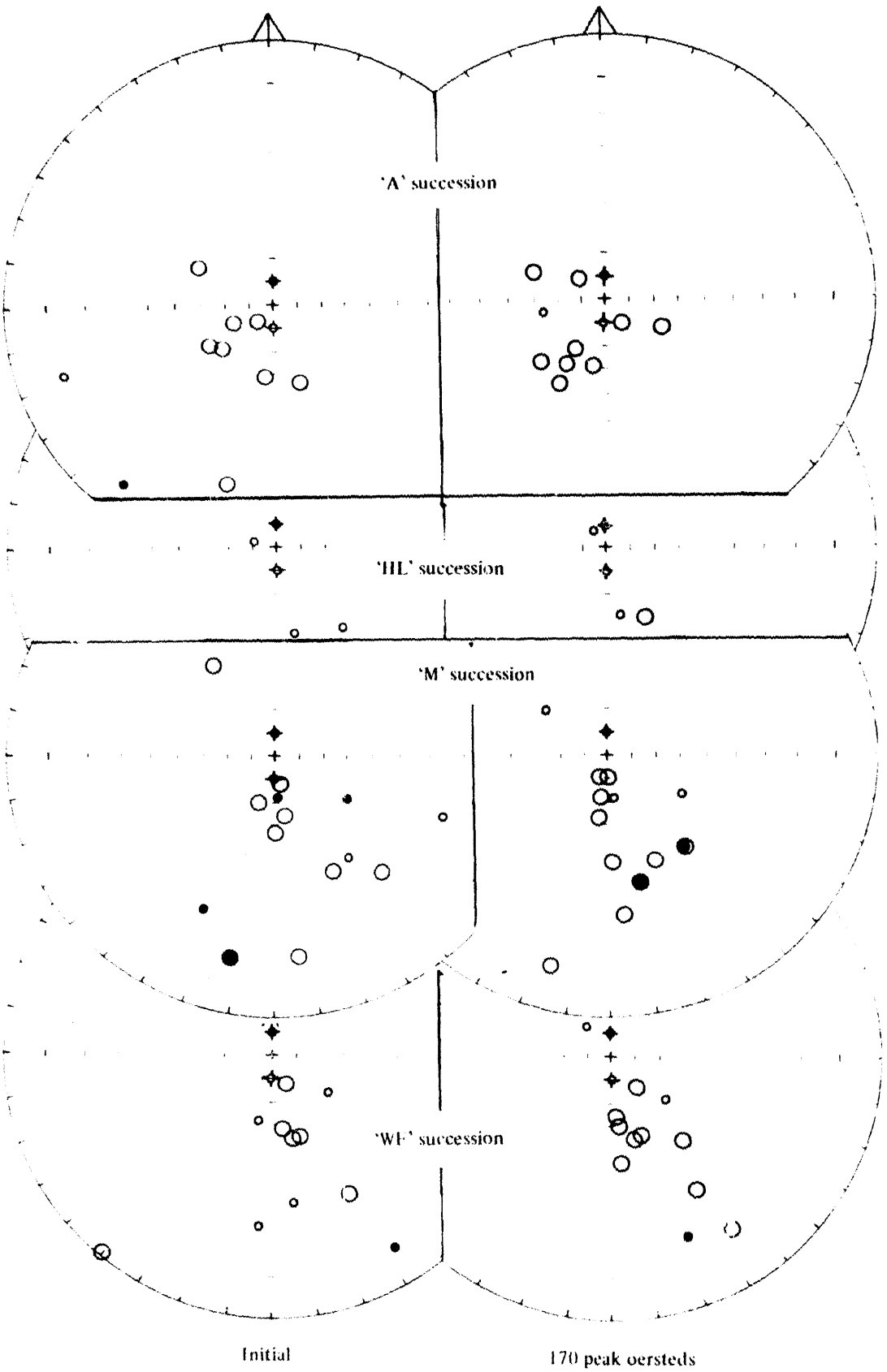


Fig. 2. Site mean directions of remanence. Large circles are site mean directions which are significant at a 99% level [15, 16]. Hollow symbols are of negative, upward inclination and solid symbols are of positive, downward inclination. The projection is equal-angle (Wulff) with the horizontal as the circumference. The geocentric axial dipole field direction is shown by ◇.

Table 2
Results of partial demagnetization of pilot samples.

Sample	Decl.	Incl.	α_{95}	Most stable range (peak oersteds)	Stability index
Succession 'A'					
10a	201	−4	25	85 − 255	0.8
8a	238	−7	34	0 − 170	0.6
6a	238	+23	4	0 − 340	4.2
4a	241	+49	49	0 − 170	0.4
2d	181	+11	7	85 − 255	2.7
Succession 'HL'					
7	46	−4	3	85 − 505	6.7
Succession 'M'					
13b	208	−81	8	85 − 675	2.5
11b	17	−38	38	255 − 675	0.7
10b	183	−66	24	255 − 505	1.1
9b	96	−25	53	85 − 675	0.5
7b	175	−22	9	85 − 255	2.3
6b	57	−36	22	255 − 505	1.1
4b	124	−47	7	0 − 340	2.4
1e	219	−67	13	0 − 285	1.5
0b	122	+31	39	85 − 675	0.6
Succession 'WF'					
L1	155	−45	1	85 − 255	14.9
J1	325	−75	37	85 − 505	0.6
H1	168	−54	2	85 − 255	11.5
F1	196	−39	43	85 − 255	0.5
E1	163	−28	5	0 − 505	3.9
C1	121	+35	8	340 − 675	3.5
A1	153	−54	2	0 − 255	9.0

Decl., Incl., are the declination and inclination of the most stable direction isolated during partial demagnetization up to 675 (peak) oersteds. α_{95} is the radius of 95% confidence about the most stable direction [17] isolated over the most stable range. Stability index [14] is the maximum value of (range ½/c.s.d.) for three or more successive measurements of the remanent direction during step demagnetization.

nents, the directions are mostly southerly with high negative inclinations.

The main exception to this generalization is that site 0 has southerly declination and positive, downward inclination. This may reflect that the polarity is changing from normal to the reversed direction observed in higher flows, or it could reflect the instability observed in the demagnetization of the pilot sample (0b). This succession can be regarded, therefore, as completely reversed although affected by instability in a few sites.

6. SUCCESSION 'WF'

The initial directions of remanence (table 1) are mostly southerly and of negative inclination, but five of the twelve sites contain widely scattered sample directions. Reduction of the low coercivity components reduced the within-site scatter in some sites, but increased it in others, so that although the mean directions remained southerly and of negative inclination, three sites contained random sample directions. The pilot samples showed a variety of behav-

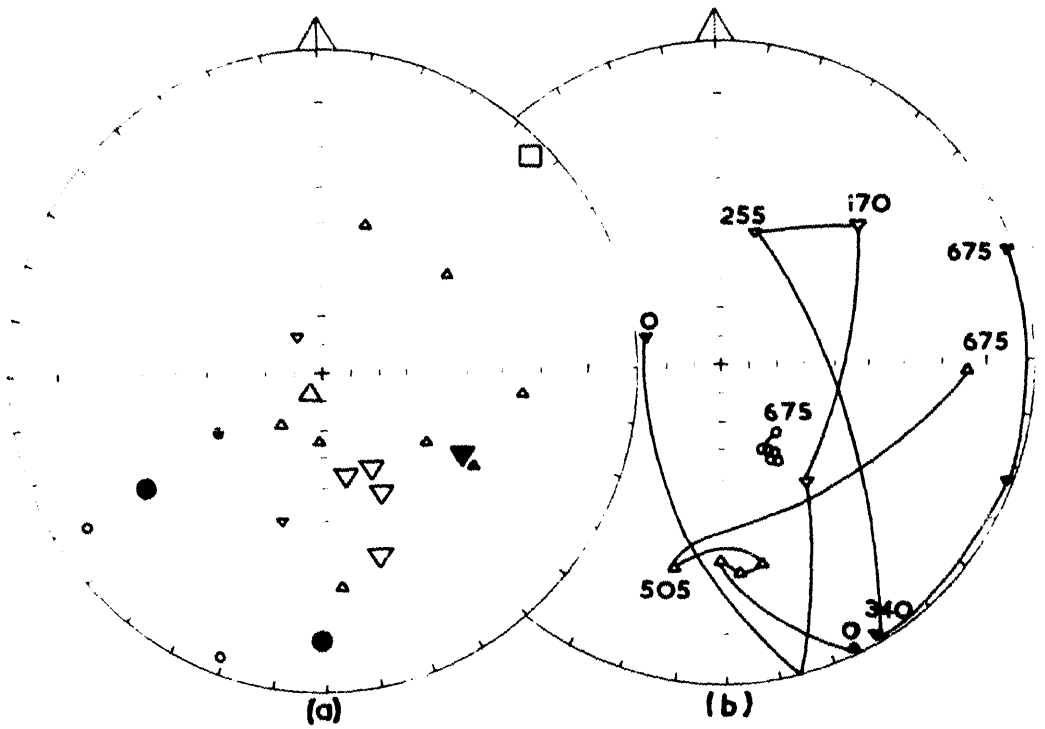


Fig. 3. Behaviour of pilot samples during partial demagnetization.

(a) Most stable directions of remanence [14] shown as large symbols if the stability index is greater than 2.5. Solid symbols are of positive downward inclination, and hollow symbols are of negative inclination.

○ 'A' succession △ 'III.' succession

△ 'M' succession ▽ 'WF' succession

(b) Three examples of directional changes in pilot samples.

○ L1 △ M7b ▽ M9b

The statistics for each example are recorded in table 2. Connecting lines do not necessarily represent the actual path of the vector between successive observations. Numbering is given in peak oersteds of applied alternating field.

Table 3

Succession mean remanent values.

Succession	M	X	No. of flows	Uncorrected		Corrected for tilt		α_{95}	Pole position	
				Decl.	Incl.	Decl.	Incl.		Lat.	Long.
A	39	16	9	211	−71	205	−67	15	66S	66W
III	31	18	1	154	−56	(147	−63)	-	(61S	24E)
M	17	10	10	174	−46	174	−50	24	51S	14W
M*	-	-	9[1]	175	−53	175	−57	18	57S	15W
WF	26	27	9	157	−50	154	−55	13	54S	9E

M, X are the geometric mean intensity and susceptibility in gauss $\times 10^{-4}$ and gauss oersted$^{-1} \times 10^{-4}$. Decl., Incl., are the mean declination and inclination calculated on all sites in which the remanent direction is significant at a 99% level after partial demagnetization in 170 (peak) oersteds. α_{95} is the radius of a cone of 95% confidence about the observed mean [17]. The pole position is calculated from the mean direction assuming an axial geocentric dipole field.

* Excluding site 0.

Table 4
Palaeomagnetic results from the early Tertiary North Atlantic Province.

	Decl.	Incl.	α_{95}	Pole position		Comment
				North	East	
British Isles	186	-63	4	78	153	1
Faeroes	176	-69	6	80	159	2
S.W. Greenland (T.D.)	160	-62	8	68	174	3
E. Greenland (a)	169	-59	17	59	172	4
(b)	168	-62	15	63	174	4
(c)	172	-62	15	69	199	4

1. Pole given by Irving [9] from 11 locality mean values (pole 11.026).
2. Abrahamsen [20] calculated on 33 specimens of basalts of 50-60 m.y. age.
3. Ketelaar [3] calculated on 15 'Trap Diabase' basalts.
4. Calculated on 4 successions: (a) before correction for tilt, (b) after correction for tilt, and (c) after rotation and translation of Greenland against an unmoved Europe on the best geometric fit [18].

iour, with two samples (F1 and J1) as metastable, ranging up to A1, H1 and L1 which were very stable.

The remanence in sites with well grouped sample directions is all of reversed polarity and similar to that observed in the 'M' succession.

7. CONCLUSIONS

7.1. *Similarity of successions*

All samples are of reversed polarity but the exact similarity of the directions cannot be assessed easily because of the variable stability of remanence. However, the locality means (table 3) of successions WF and M are very similar, and their circles of 95% confidence overlap with that of the succession 'A' mean. The separate localities may, therefore, be treated as having similar directions of remanence and the overall mean (table 4) can be estimated with the most conservative statistical limits by giving equal weight to each locality mean. Calculation of this overall mean before and after correction for the small amount of tectonic tilt at each locality shows the mean direction to be more precisely defined after correction for tilt, suggesting that the tilt occurred after the formation of the lavas. However, the tilts involved are small and make little difference to the actual mean direction or to its precision parameters.

7.2. *Reversals*

All samples which are stable have reversed directions, but it is not known what time span represented by the different sequences. However, it is tempting to suggest that sequences belong to the same lengthy polarity period as has been observed in the Antrim Basalts [19] and Faeroe Islands [20]. However, radioactive age determinations are currently in progress which will establish the actual age of the successions, thereby allowing correlations with other rocks of similar age to be made.

7.3. *Secular variation*

An estimate of secular variation cannot be made on these lavas as the magnetic stability is considered to be too variable, and generally too low for the precise determinations of the earth's magnetic field required for this type of study.

7.4. *Directions of remanence and continental drift*

The mean directions of remanence are displaced significantly from the reversed axial geocentric dipole field (180,-79) in this area. This displacement may be due to polar wandering alone or polar wandering plus continental drift. Unfortunately the age is not known sufficiently accurately to estimate the displacement attributable to polar wandering, but it is of interest to compare these results with the mean directions determined in other rocks of the North Atlantic Tertiary Province (table 4). The pole positions for the Trap Diabase rocks of S.W. Greenland and those of the East Greenland Tertiary lavas are very similar suggesting contemporaneity, but both differ from the pole

positions determined for the British Isles and the Faeroe Islands. It is difficult to establish the significance of this difference as the published statistics have been calculated giving equal weight to different levels of measurement. However, it is thought that there is a real difference between the Greenland and European results. This difference is somewhat increased if Greenland is rotated against Europe in the position envisaged by Bullard, Everett and Smith [18].

ACKNOWLEDGEMENTS

The author wishes to thank members of Oxford University, on both expeditions, for their co-operation in collecting oriented samples; in particular Drs. J. C. Rucklidge and C. K. Brooks. The assistance of Miss A. Wrightson and members of staff of the Department of Geophysics and Planetary Physics of the University of Newcastle upon Tyne is gratefully acknowledged. The work was carried out as part of a N.E.R.C. research project awarded to Professor S. K. Runcorn.

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